

Synthesis of PLA by using ionic liquids as catalysts

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Abstract

The poly(lactic acid) – PLA is a versatile polymer material widely used in different fields, including technological and biomedical applications. PLA is a biodegradable and bioabsorbable material, whose synthesis is relatively easy. Due to these characteristics, several researches are performed in an attempt to cheapen the process and allowing its diffusion. Some synthesis routes were used in this paper to produce lactide monomer from lactic acid and subsequently, the ring-opening polymerization using ionic liquid catalysts to obtain PLA. The reactions were carried out in three stages: *i*) synthesis of prepolymer from L-lactic acid through polycondensation; *ii*) synthesis of lactide from prepolymer; *iii*) PLAs production from lactide by using ionic liquid catalysts: BMI.In₂Cl₇, BMI.Cu₂Cl₃ and BMI.Sn₂Cl₅. Ring-opening polymerizations – (ROP) were also carried out employing the above-mentioned Lis catalysts, varying the temperature and time of synthesis and the monomer lactide/catalyst molar ratio. The polymeric materials were characterized by FTIR, NMR and GPC. The preliminary results showed that ionic liquid catalysts have a great potential for synthesis of PLAs.

Keywords: Lactic acid, poly(lactic acid), ionic liquids, ring-opening polymerization

1 Introduction

The biodegradable polymer poly(lactic acid) – PLA belonging to the family of linear polyester are widely studied due to its low environmental impact, besides being one of the most important promising, due to thermal and mechanic properties for replacing the conventional polymers which are made from petrochemical origins. Its features such as biocompatibility biodegradability, and sustainability which is makes quite sought after for industries, even though, its applications to be limited for low toughness and slow crystallization rate [1, 2].

PLA can be obtained by polycondensation of lactic acid, which is derived from sustainable sources. However, low molar mass PLA is usually obtained, due to the existence of chemical equilibrium in the condensation reaction and the resulting depolymerization reactions. Poly(lactic acid), industrially is synthesized by ring-opening polymerization (ROP) of lactide, a cyclic ester of lactic acid. The cationic polymerization is a technique widely used to produce polymers by ROP. The cationic polymerization occurs in the Lewis acids presence, electrophiles, and depending on of reaction conditions, cationic polymerization can be whether SN1 or SN2 [3, 4].

The cationic polymerization Sn2 occurs when the electrophile is attacked for whether atom or by functional group Y (Lewis Base) and thereby, Y become a cationic ion, due to electron pair donated to electrophile. Consequently, Y will attract bonding electrons with X atom through the coulomb force, as result generating a X cation ion that it will receive electron from the next Y atom belonging the monomer and thus reaction occurs successively forming polymeric chain [5-7].

The process of cationic polymerization of type Sn1 works the same way as Sn2, however the only difference is that in Sn1, occurs firstly the ring opening polymerization (ROP) which will produce an opening chain cationic monomer (electrophile) that will be attacked by Y atom (nucleophile) of neighbor monomer and so on producing active point of growing chain until its formation in a polymeric chain, while that Sn2 has firstly a formation of cationic monomers cyclic which is attacked by neighbor monomers [8].

The cationic ring-opening polymerization of lactide is induced many times for using organic acid catalysts, but this paper it was replaced by ionic liquids. The reaction gets starting with activation of monomer for protonation and subsequently the attack of carbonyl group of the initiator alcohol in lactide ring, and then forming lactil alcohol. The deprotonation of lactil alcohol cause the breaking of ring and consequently, the group terminal hydroxyl will react with lactide monomers and inducing the breaking of ring itself and so on.

Halogenated alkanes are highly used as solvents during the preparation of traditional cationic polymerization. The main issue about using halogenated alkanes as solvent are cause environments pollution, by the fact they were quite volatile, and toxic. In addition, ionic liquid can be regarded as the ideal replacing to halogenated alkane solvents. Apart from being no pollution to the environment and recycled, they are also show low energy consumption [9].

In this study, catalysts of imidazolium-based ionic liquids incorporated with different metals were successfully used in cationic ring-opening polymerization of lactide. We also investigated the effects of various reaction parameters, such as temperature, reaction time and lactide/catalyst molar ratio.

1. Materials and methods

1. Materials

Polymerization reactions were conducted using lactide (Sigma-Aldrich, purity $\geq 99\%$) in the presence of ionic liquids catalysts 1-butyl-3-methylimidazolium trichlorodocuprate(I) (BMICu₂Cl₃), pentachlorotinate(II) (BMI.Sn₂Cl₅) and heptachlorodiindate(III) (BMI.In₂Cl₇), synthesized as reported by Dutra et al. [10] or dibutyltin dilaurate (DBTL, Sigma-Aldrich, purity $\geq 95\%$) in inert atmosphere of nitrogen gas (N₂) industrial 4.5 (White Martins, purity $\geq 99.99\%$).

1. Synthesis of poly(lactic acid) – (PLA) using ionic liquids and commercial catalysts

Ring-opening polymerization of lactide reactions were carried out at different reaction temperatures (110, 120, and 130 °C) under magnetic stirring (xx rpm) and inert atmosphere of nitrogen gas in a Schlenk flask. Initially, the catalysts (BMICu₂Cl₃, BMI.Sn₂Cl₅, BMI.In₂Cl₇ or DBTL) were weighed in Schlenk tube, using a catalyst/monomer molar ratio varying of 1:800, 1:650 and 1:500. Then, lactide (about of xx g) was added and, after reaching the reaction temperature, the system was maintained for 24 h under constant agitation.

After completion of the reaction, the mixture was cooled, and dichloromethane was added to stop polymerization. The solvent was evaporated and then placed in the oven at 60 °C for approximately 48 h. The dried samples were solubilized in dichloromethane and vacuum filtered using a chromatographic column packed with Celite 545 P.A. (Synth) and a thin layer of basic alumina to remove the catalyst.

1. Material characterization

Nuclear magnetic resonance (^1H -NMR) spectra were obtained using a Bruker Ascend 600 (14 T) spectrometer, equipped with a 5 mm probe operating at 600 MHz. About 20 mg of the catalysts of ionic liquids were dissolved in 0.5 mL of deuterated water (D_2O , Sigma-Aldrich, purity $\geq 99.9\%$), while the synthesized poly(lactic acid) were dissolved in deuterated chloroform (CDCl_3 , Synth, purity $\geq 100\%$) and the spectra were acquired expressing chemical shifts in parts per million (ppm).

The differential scanning calorimetry (DSC) curves were obtained using a Shimadzu model DSC-60 equipment. The initial masses used were approximately 5.0 mg and aluminum crucibles were used. The measurements were made under atmosphere of He at a flow rate of 30 mL min^{-1} , with heating rate of $10 \text{ }^\circ\text{C min}^{-1}$ and two heating ramps (-40 to $180 \text{ }^\circ\text{C}$). The second heating cycle was used to determine the melting temperature (T_m), melting enthalpy (H_m), crystallinity (X_c) and glass transition temperature (T_g) of the polymers.

The mass-average molar mass (M_w) and the number-average molar mass (M_n) and the dispersity (ĐM) of the polymers were determined using a gel permeation chromatograph (Malvern, model Viscotek RImax) equipped with a refractive index detector with a set of three columns of $300 \times 8 \text{ mm}$ mounted in series (KF-802.5, KF-804L and KF-805L) operating at $40 \text{ }^\circ\text{C}$. The system was calibrated using polystyrene standards with molar mass ranging from 1.20 kg mol^{-1} to 4500 kg mol^{-1} and monodisperse (ĐM close to 1.0). Tetrahydrofuran solvent (THF) (HPLC grade, Sigma-Aldrich, purity $\geq 99.99\%$) was used as the mobile phase with a flow rate of 1 mL min^{-1} . The previously prepared solutions ($1.5 \text{ mg sample}/1.0 \text{ mL THF}$) were filtered through hydrophobic polytetrafluoroethylene (PTFE) membranes with $0.45 \text{ }\mu\text{m}$ pore size and injected into the equipment with a volume of $100 \text{ }\mu\text{L}$.

FTIR

Conversão

1. Result and discussion

• Characterization of ionic liquids

The ^1H -NMR spectra of BMI.Cl, BMI. In_2Cl_7 , BMI. Cu_2Cl_3 and BMI. Sn_2Cl_5 were analyzed and the signals were identified and shown in Figure 1.

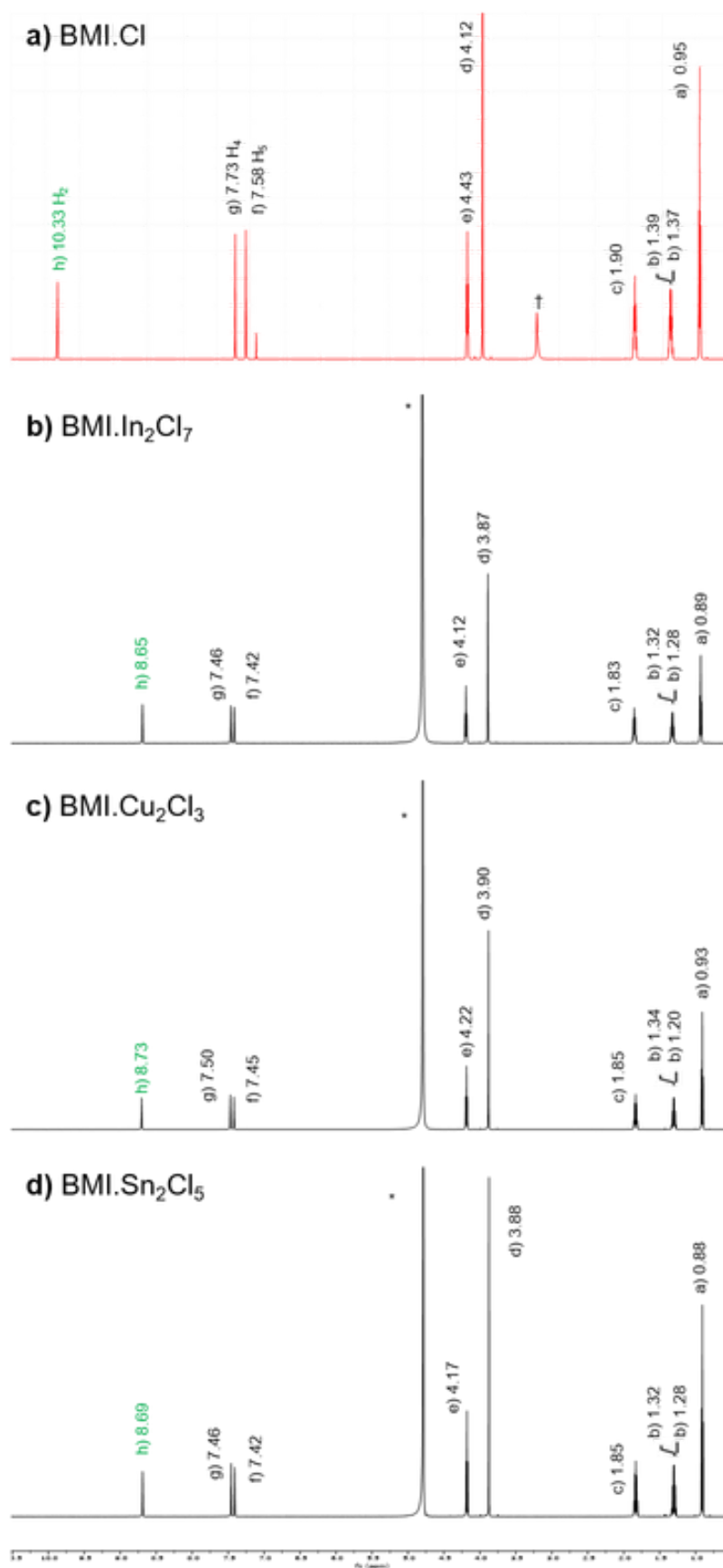


Figure 1: ¹H-NMR spectra of ionic liquids a) BMI.Cl obtained in DMSO; b) BMI.In₂Cl₇, c) BMI.Cu₂Cl₃ and d) BMI.Sn₂Cl₅ obtained in D₂O.

Using BMI.Cl spectrum as patterns to identify the signals the following shifts can be confirmed: h: 0.91 () ppm (3H), g: 1.31 () ppm (2H), f: 1.86 ppm (2H), a: 3.8 () ppm (3H), e: 4.8 () ppm (2H), d: 7.40 ppm () (1H), c: 7.46 ppm () (1H) and b: 8.69 () ppm (1H). By the fact of hydrogen b: 8.69 ppm (1H) to be extremely sensible due to anions around itself and because of that its signals are a bit more shifted if compared from the rest [11]. †

Figure 2 shows the Fourier-transform infrared spectrum (FTIR) of the ionic liquids BMI.Cl, BMI.In₂Cl₇, BMI.Cu₂Cl₃ and BMI.Sn₂Cl₅.

Figure 2: FTIR spectra of synthesized ionic liquids BMI.Cl, BMI.In₂Cl₇, BMI.Cu₂Cl₃ and BMI.Sn₂Cl₅.

- **Characterization of lactide and poly(lactic acid)**

The monomer, lactide, was characterization from ¹H-NMR and the spectrum shown in Figure 3. The signals between = 5.06 - 5.03 ppm are assigned to -CH₃ and consequently the shifts between 1.67 - 1.65 ppm are represented for -CH of the cyclic ester.

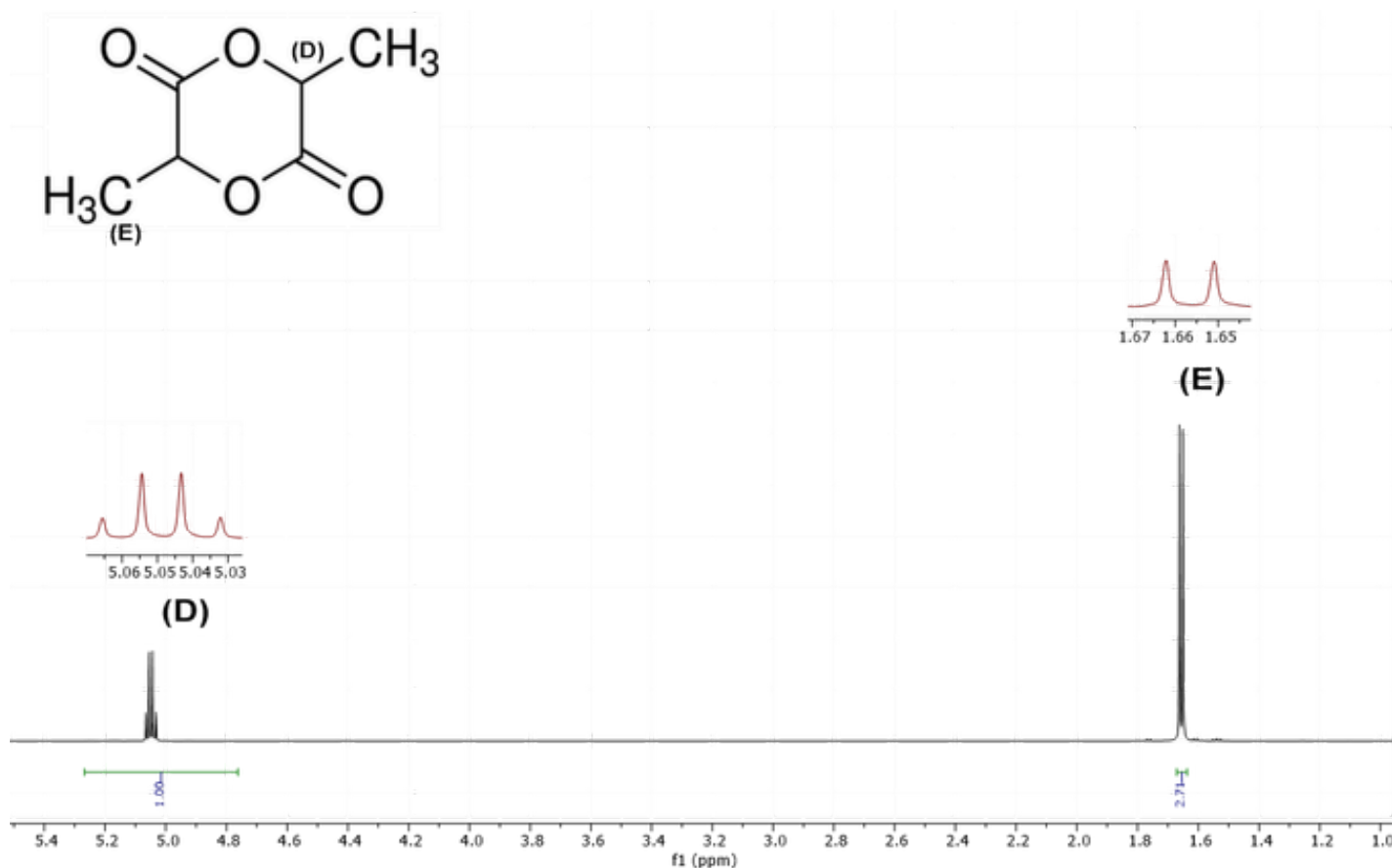


Figure 3: ¹H-NMR spectrum of lactide monomer obtained in CDCl₃.

The polymers poly(lactic acid) – PLA produced from using ionic liquids catalyst in overall showing features as solids, brittle and with different coloring in according with kind of ionic liquid used for synthesizing. However, the PLA produced using BMI.Cu₂Cl₃, and dibutyltin dilaurate (DBTL) showed distinctive features such as semi-liquid material, and pasty, a bit different from PLA synthesized with using BMI.In₂Cl₇ and BMI.Sn₂Cl₅.

The Figures 3, 4, 5 and 6 shows the FTIR spectra of PLA using different catalysts. The bands between 3520 cm⁻¹ and 3505 cm⁻¹ refer to stretching (O-H) in the terminal chain. Furthermore, the two of bands observed between 3009 cm⁻¹ and 2935 cm⁻¹ referents about symmetric and asymmetrical stretching of C-H sp³ (methyl group) and the bands observed in 1465 cm⁻¹ and 1360 cm⁻¹ can be assign, respectively, for angular deformation of methyne (-CH) and methyl groups (-CH₃), indicating the formation of the polymeric chain of the PLA, through the opening of the lactide ring. The carbonyl (C=O) signal can be notice between 1758 cm⁻¹ and 1750 cm⁻¹. The peak 1170 cm⁻¹ is assigned for the symmetric stretching C-O of ester group present in the polymeric structure.

The result observed in the table 11 is realized the shifting of band C-O-C assigned about ester, which for lactide is 1155 cm⁻¹ and for poly (lactic acid) – (PLA) is 1190 cm⁻¹, this displacement is the confirmation about the synthesizing and the opening ring mechanism.

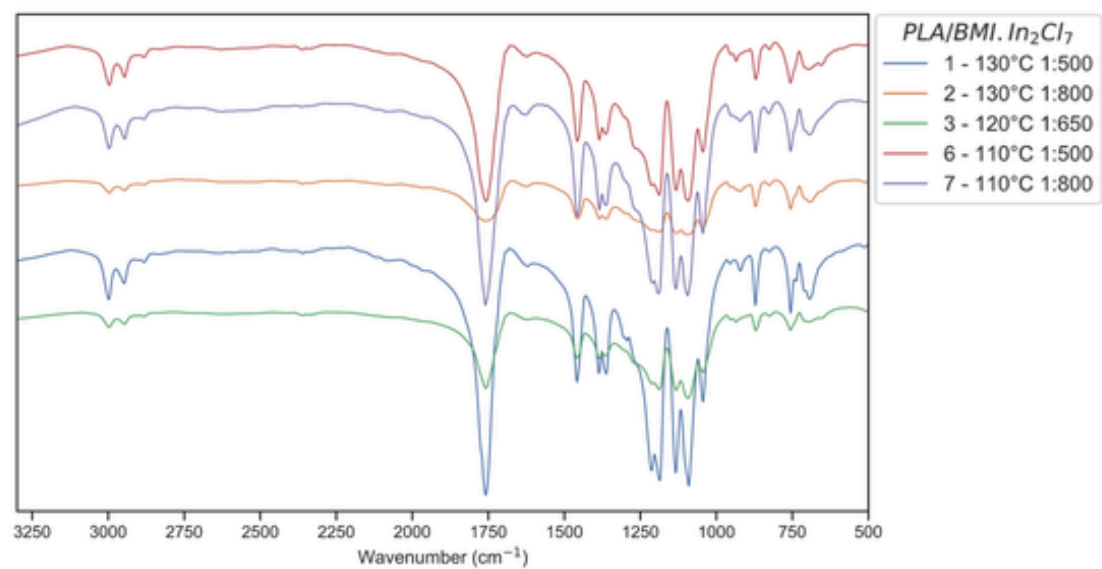


Figure 3. Fourier transform infrared (FTIR) spectra of poly (lactic acid) by using $\text{BMI} \cdot \text{In}_2\text{Cl}_7$.

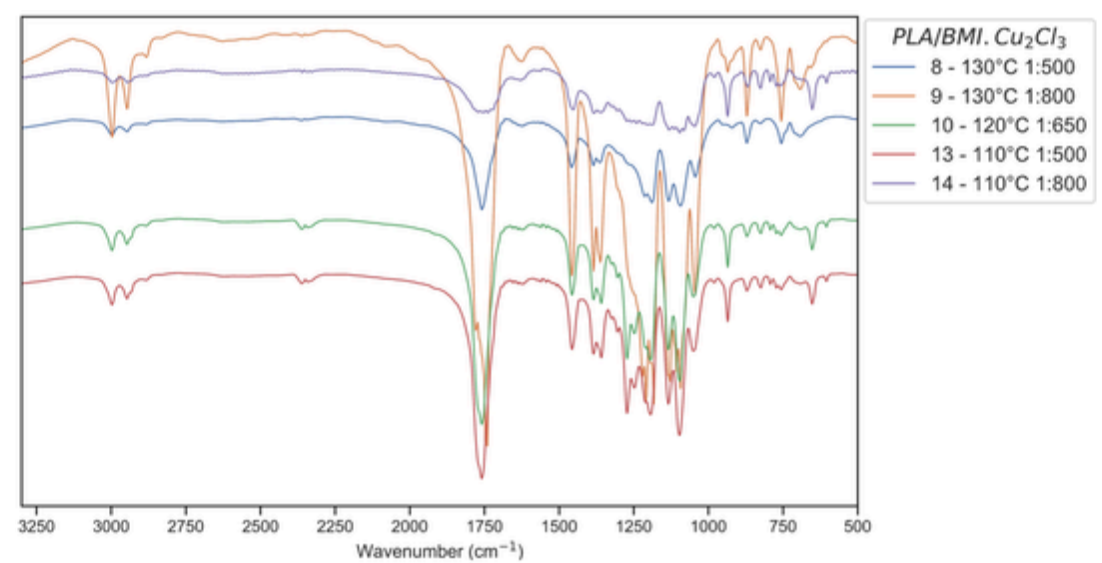


Figure 4. Fourier transform infrared (FTIR) spectra of poly (lactic acid) by using $\text{BMI} \cdot \text{Cu}_2\text{Cl}_3$.

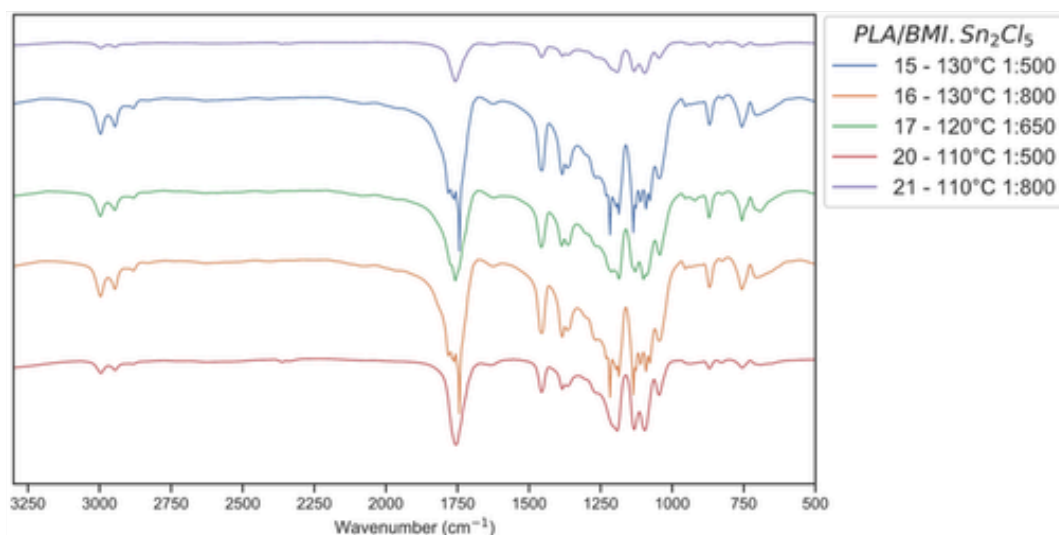


Figure 5. Fourier transform infrared (FTIR) spectra of poly (lactic acid) by using BMI.Sn₂Cl₅.

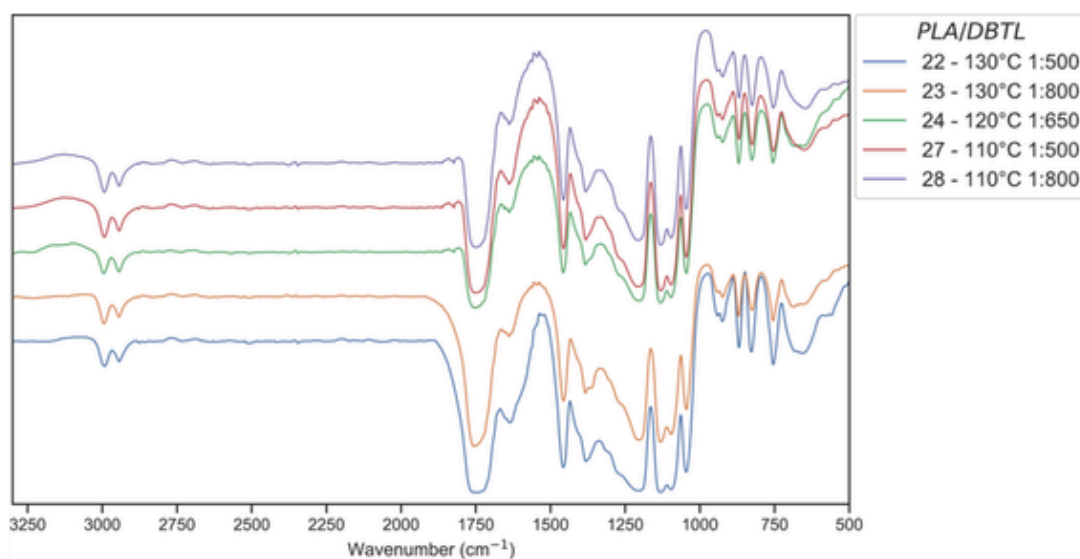
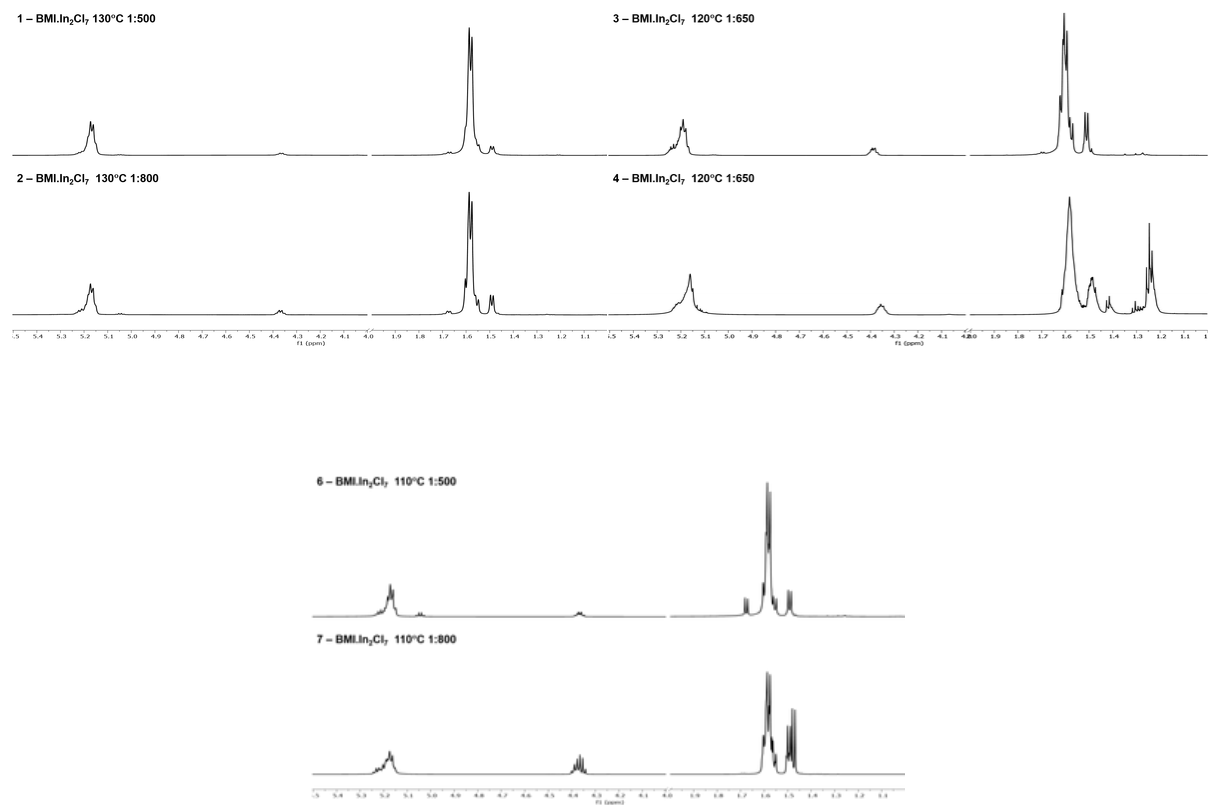


Figure 6. Fourier transform infrared (FTIR) spectra of poly (lactic acid) by using DBTL.

In ¹H-NMR spectrum of the PLA synthesized, Figures 7 - 10, can be observed the signals around 4,38 - 4,4 () ppm referents about the chain end (methyne end) =CH-, and also together with the signals 5,1 ppm which is assigned to (methyne) =CH-, the signal 1,57 ppm belonging about group (methyl) -CH₃, and finally the signal 6.5, 6.2 and 5.0 () ppm assigned to group vinyl [12]. Thus, the characterization for ¹H-NMR spectroscopy demonstrate of poly (lactic acid) – (PLA) forming through

by ring opening polymerization and so confirmed by means ¹H-NMR spectroscopy its synthesis. Overlap of signals is a feature often that it can be observed in ¹H-NMR spectrum, due to formation of small chains during the synthesis [12].



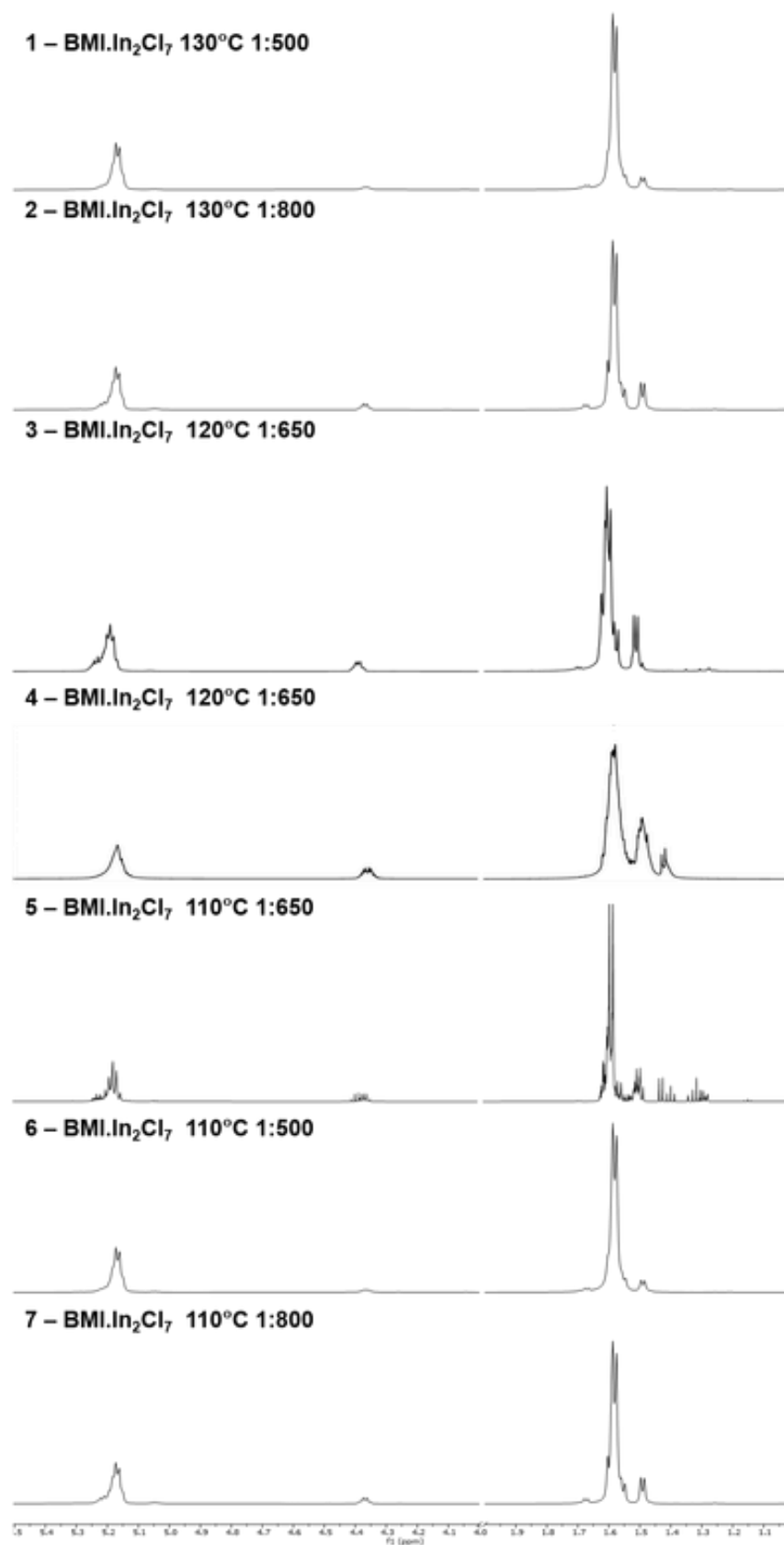


Figure 7. ¹H-NMR spectrum of poly (lactic acid) synthesized with BMI.In₂Cl₇ in CDCl₃.

8 – BMI.Cu₂Cl₃ 130°C 1:500



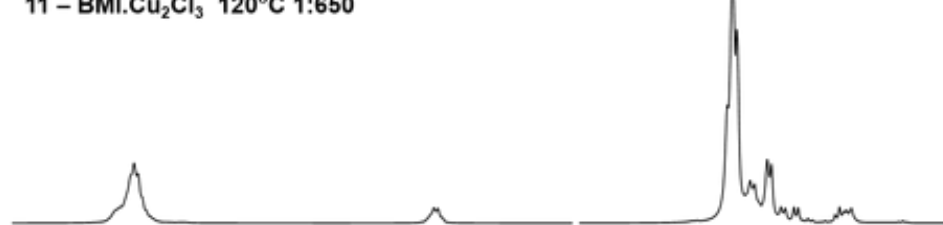
9 – BMI.Cu₂Cl₃ 130°C 1:800



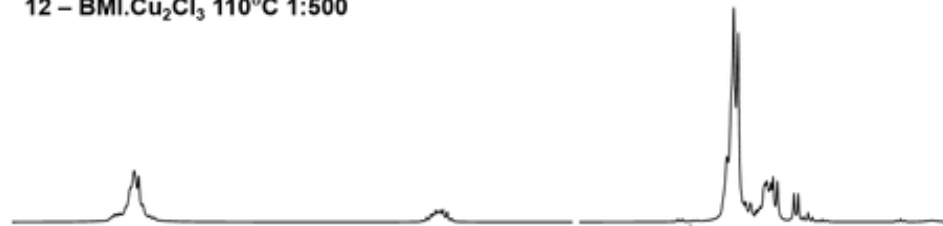
10 – BMI.Cu₂Cl₃ 120°C 1:650



11 – BMI.Cu₂Cl₃ 120°C 1:650



12 – BMI.Cu₂Cl₃ 110°C 1:500



13 – BMI.Cu₂Cl₃ 110°C 1:500



14 – BMI.Cu₂Cl₃ 110°C 1:800

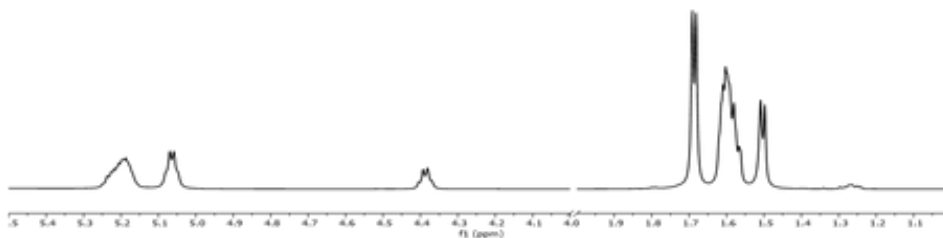


Figure 8. ¹H-NMR spectrum of poly (lactic acid) synthesized with BMI.Cu₂Cl₃ in CDCl₃.

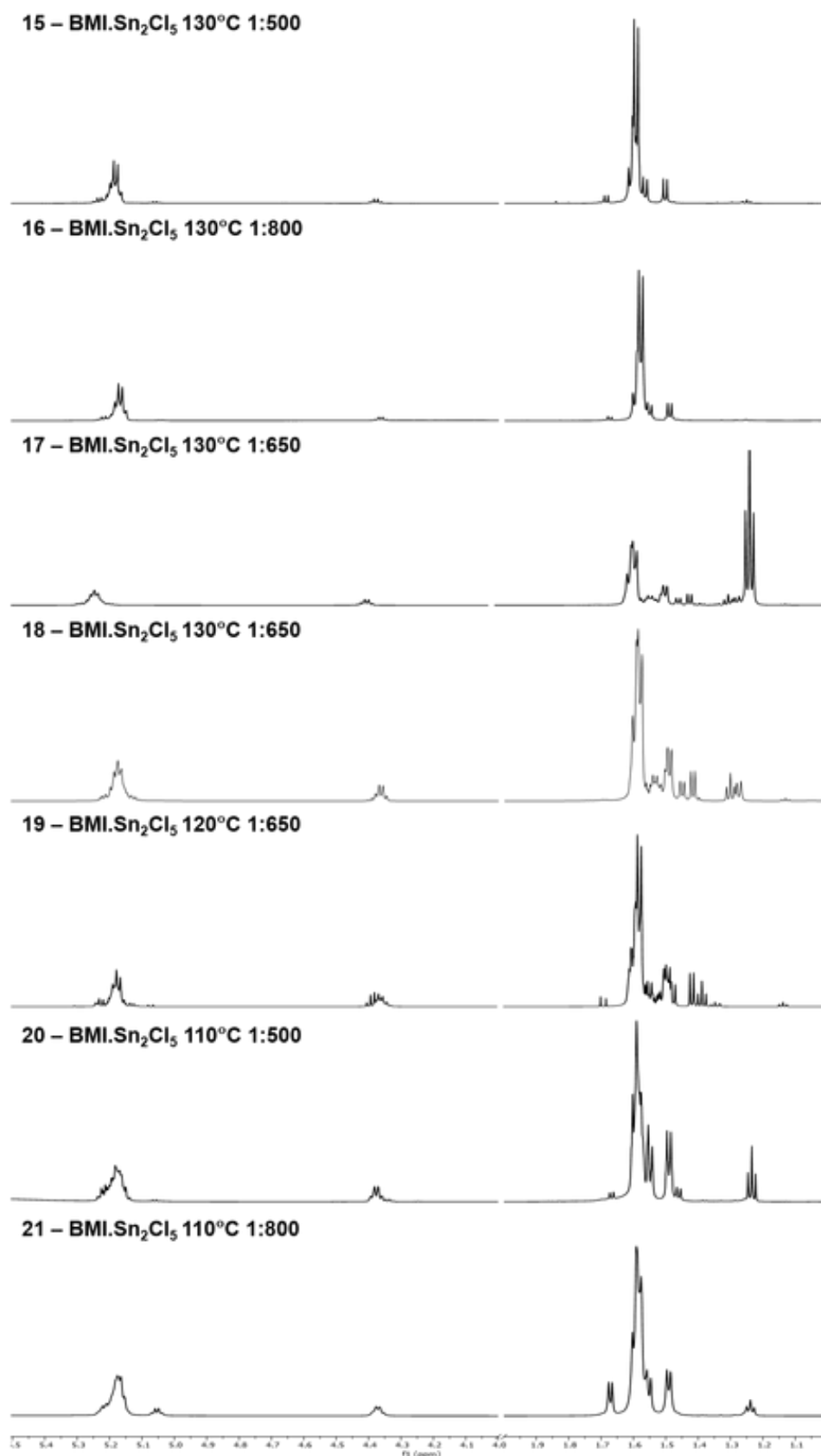


Figure 9. ¹H-NMR spectrum of poly (lactic acid) synthesized with BMI.Sn₂Cl₅ in CDCl₃.

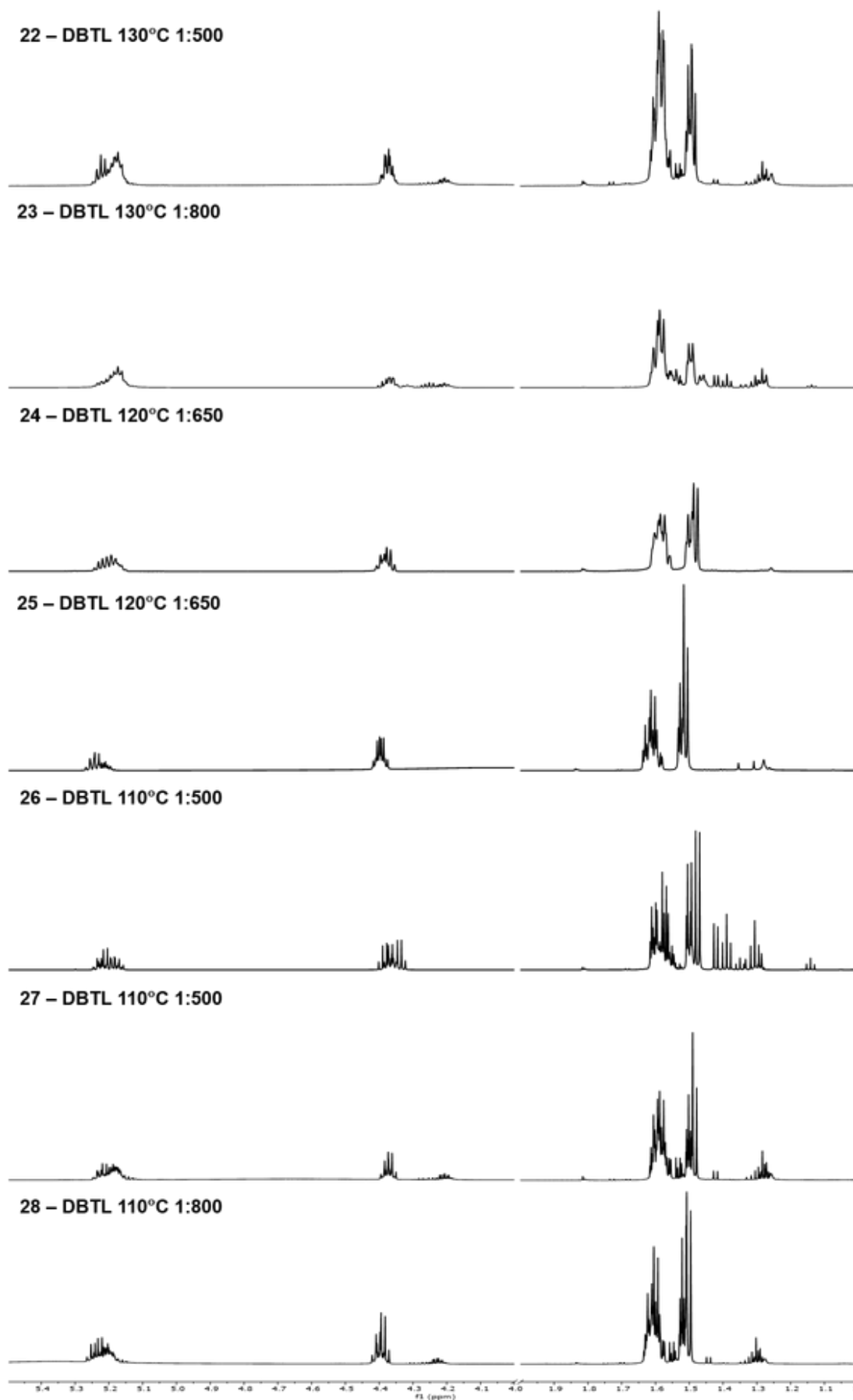


Figure 10. ¹H-NMR spectrum of poly (lactic acid) synthesized with DTBL in CDCl₃.

The conversion of polymers was obtained by the method of relative integration of the $^1\text{H-NMR}$ spectrum, using the area of **xx group of** PLA in $\delta = 5.16$ ppm divided by the sum of area of **xx group of** PLA, $\delta = 5.16$ ppm, more area of **xx group of** monomer lactide in $\delta = 5.03$ ppm, as shown in Equation 1 [13, 14].

Table 1 shows the performance of different catalysts in relation to the conversion of PLA in polymerizations performed in different temperatures and catalyst/lactide molar ratio at 24 h of synthesis.

In overall the syntheses of PLAs show with a relatively good yield, the most of them turn around 90% of conversion. Even so, the ionic liquid BMI.Cu₂Cl₃ did not show itself a good catalyst, mainly in a temperature 110 and 120 °C, since it caused the formation of polymers with 70 to 85% conversion, in addition produced a PLA with features semi-liquid and pasty.

Other issue that it was noticed, it was with relation about using the industrial catalyst dibutyltin dilaurate – (DBTL), even though showing high yield, all PLAs synthesized in the presence of this catalyst showed features quite pasty, no solid PLAs are obtained.

Table 1. Conversions of poly (lactic acid) – (PLA) synthesized with ionic liquids (ILs) and Dibutyltin dilaurate – (DBTL) at different catalyst/lactide molar ratio and temperature.

Catalyst	Entry	Molar ratio	T (°C)	Conversion (%)
BMI.In ₂ Cl ₇	1	1:500	130	98
	2	1:800		98
	3	1:650	120	98
	4			96
	5			-
	6	1:500	110	86
	7	1:800		89
BMI.Cu ₂ Cl ₃	8	1:500	130	98
	9	1:800		90
	10	1:650	120	68
	11			70
	12			-
	13	1:500	110	85
	14	1:800		68

BMI.Sn ₂ Cl ₅	15	1:500	130	98
	16	1:800		98
	17	1:650	120	92
	18			89
	19			-
	20	1:500	110	96
	21	1:800		95
DBTL	22	1:500	130	98
	23	1:800		98
	24	1:650	120	97
	25			98
	26			-
	27	1:500	110	98
	28	1:800		98

The molar mass distribution curves of all polymers synthesized are shown in Figures 11, 12, 13 and 14. The theory number average molecular weight (M_n)*, number average molecular weight (M_n), weight average molecular weight (M_w) and molar-mass dispersity ($\overline{M}_w/\overline{M}_n$) are presented in Table 2.

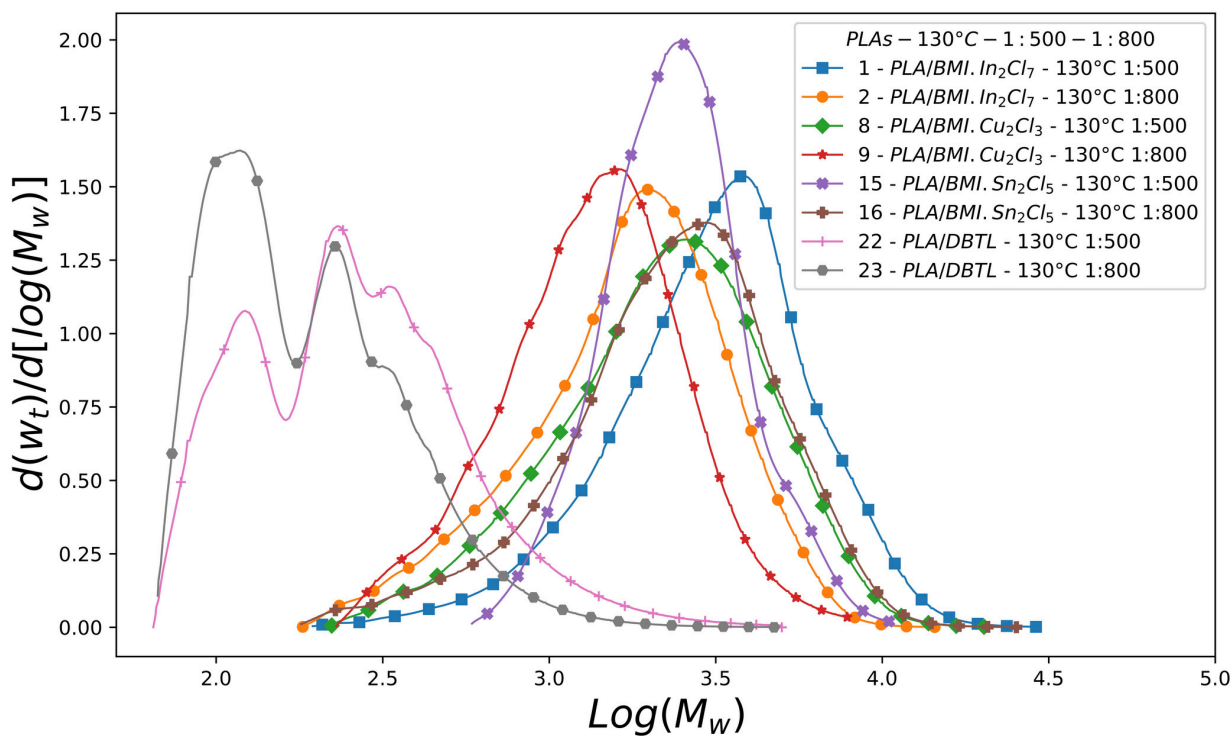


Figure 11. Molar mass distribution curves of the poly (lactic acid) – PLA synthesized by using ILs and DBTL catalysts at 130°C.

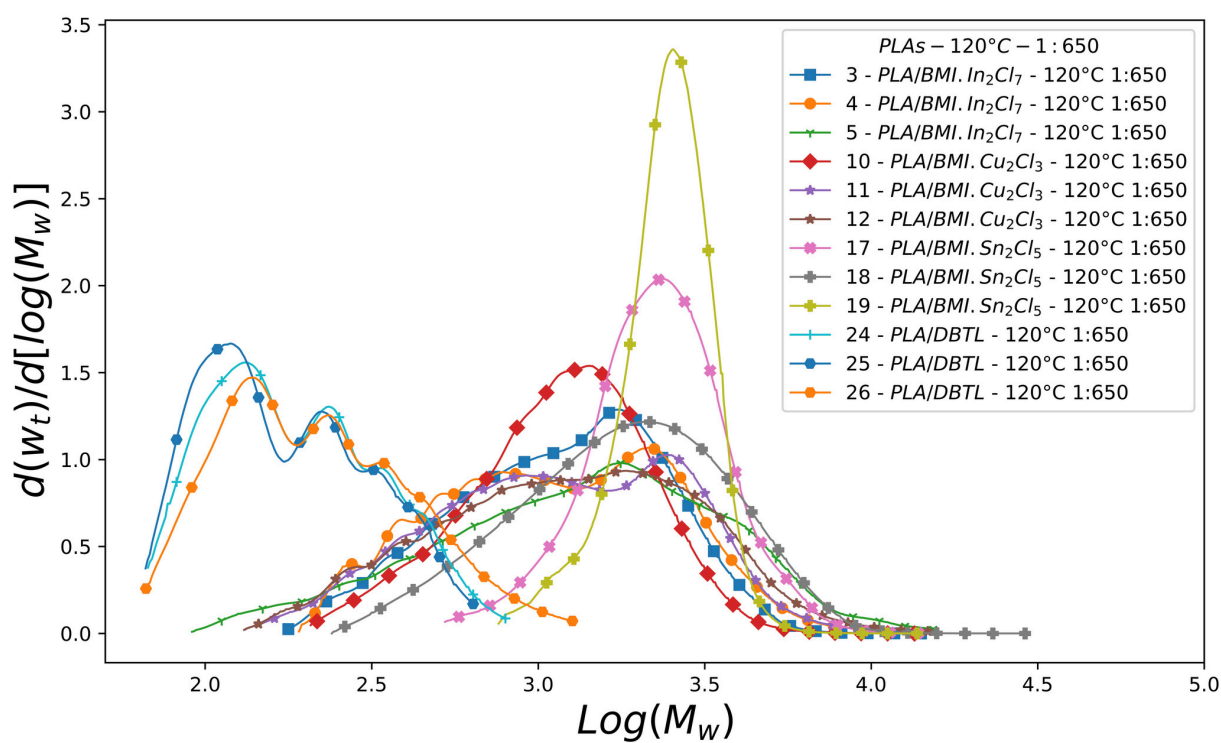


Figure 12. Molar mass distribution curves of the poly (lactic acid) – PLA synthesized by using ILs and DBTL catalysts at 120°C.

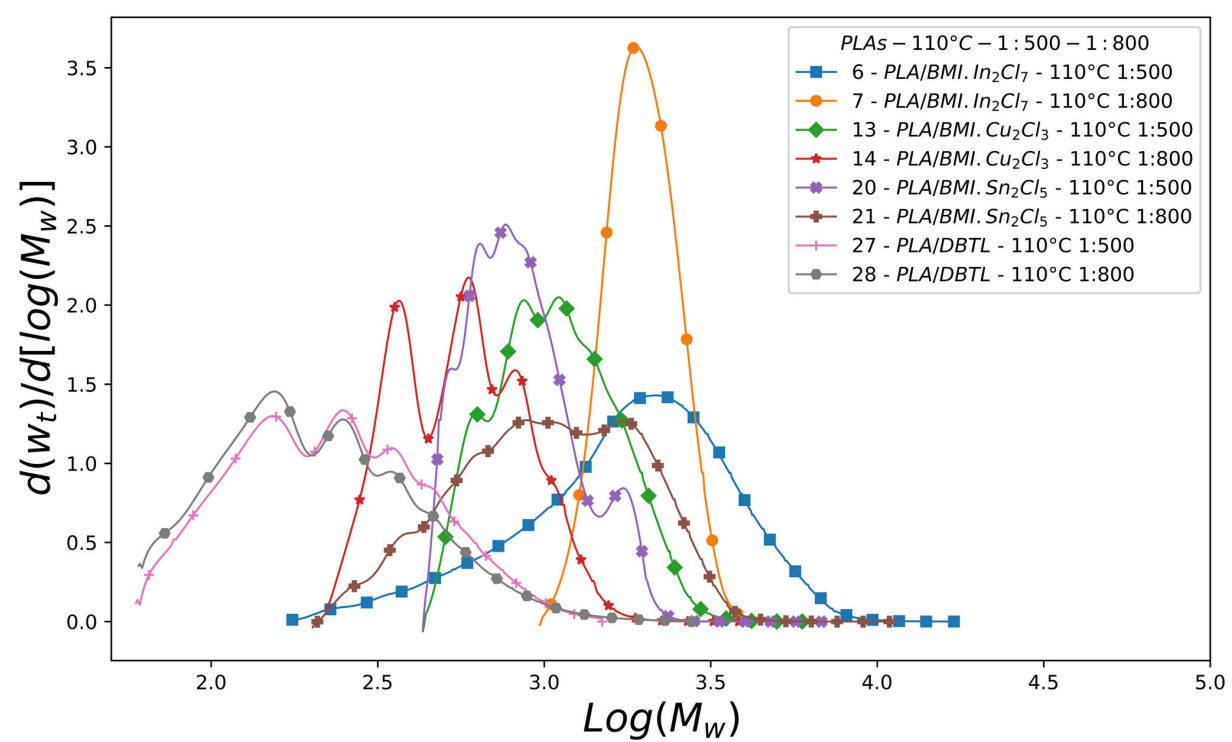


Figure 13. Molar mass distribution curves of the poly (lactic acid) – PLA synthesized by using ILs and DBTL catalysts at 110°C.

The poly (lactic acid) – (PLA) synthesized by using of dibutyltin dilaurate - (DBTL) catalyst presented polymers with and quite low, even at temperature 130°C with $\bar{M}_n = 323 \text{ kg mol}^{-1}$, $\bar{M}_w = 385 \text{ kg mol}^{-1}$, comparing with PLAs produced by using ionic liquids catalysts. However, it seems low values to $\bar{M}_n$ and $\bar{M}_w$ are sort of features to PLAs produced by using DBTL, because in according D.K. Yoo et al [15] even using reaction conditions at temperature 180°C, it was produced $\bar{M}_n$ and $\bar{M}_w$ with values around 730 kg mol⁻¹ and 650 kg mol⁻¹, respectively.

Table 2. Average molar masses and molar-mass dispersity of the polymers synthesized at 130°C, 120°C, and 110°C using ILs catalysts and DBTL commercial catalyst.

Entry	Catalyst	(theory)(kg mol-1) *	(kg mol-1)	(kg mol-1)	DM
1	BMI.In2Cl7	1512.00	2450	3706	1.51
2	BMI.In2Cl7	872.00	1498	2207	1.47
3	BMI.In2Cl7	726.54	1571	1805	1.14
4	BMI.In2Cl7	972.50	1497	2070	1.38
5	BMI.In2Cl7	-	1515	2130	1.40
6	BMI.In2Cl7	872.01	1196	1884	1.57
7	BMI.In2Cl7	625.84	818	1347	1.11
8	BMI.Cu2Cl3	1872	2172	2426	1.11
9	BMI.Cu2Cl3	1512	1625	2745	1.68
10	BMI.Cu2Cl3	1872	1105	1617	1.46
11	BMI.Cu2Cl3	972	810	1603	1.97
12	BMI.Cu2Cl3	-	1270	2020	1.59
13	BMI.Cu2Cl3	1272	987	1167	1.18
14	BMI.Cu2Cl3	672	509	614	1.20

15	BMI.Sn2Cl5	2400	2115	2654	1.25
16	BMI.Sn2Cl5	1100	1671	2901	1.73
17	BMI.Sn2Cl5	110.57	2002	2514	1.255
18	BMI.Sn2Cl5	972	1275	1741	1.36
19	BMI.Sn2Cl5	-	2310	2532	1.10
20	BMI.Sn2Cl5	625.84	803	917	1.14
21	BMI.Sn2Cl5	522.00	841	1223	1.45
22	DBTL	348.92	323	385	1.19
23	DBTL	290.18	300	352	1.17
24	DBTL	192.30	249	257	1.03
25	DBTL	216.00	274	297	1.08
26	DBTL	-	271	283	1.04
27	DBTL	432.00	275	284	1.03

*The theory Molecular Weight Averages were calculated using relation spectrum 1H-NMR between 5.18 ppm and 4.36 ppm.

Through the gel permeation chromatography of synthesized PLAs, it can be realized that some of the samples of PLA/BMI.Cu2Cl3, and PLA/DBTL shows formation of micro molecules with different sizes (oligomers), which is showed by means of different types of curves in Figure X [16-19].

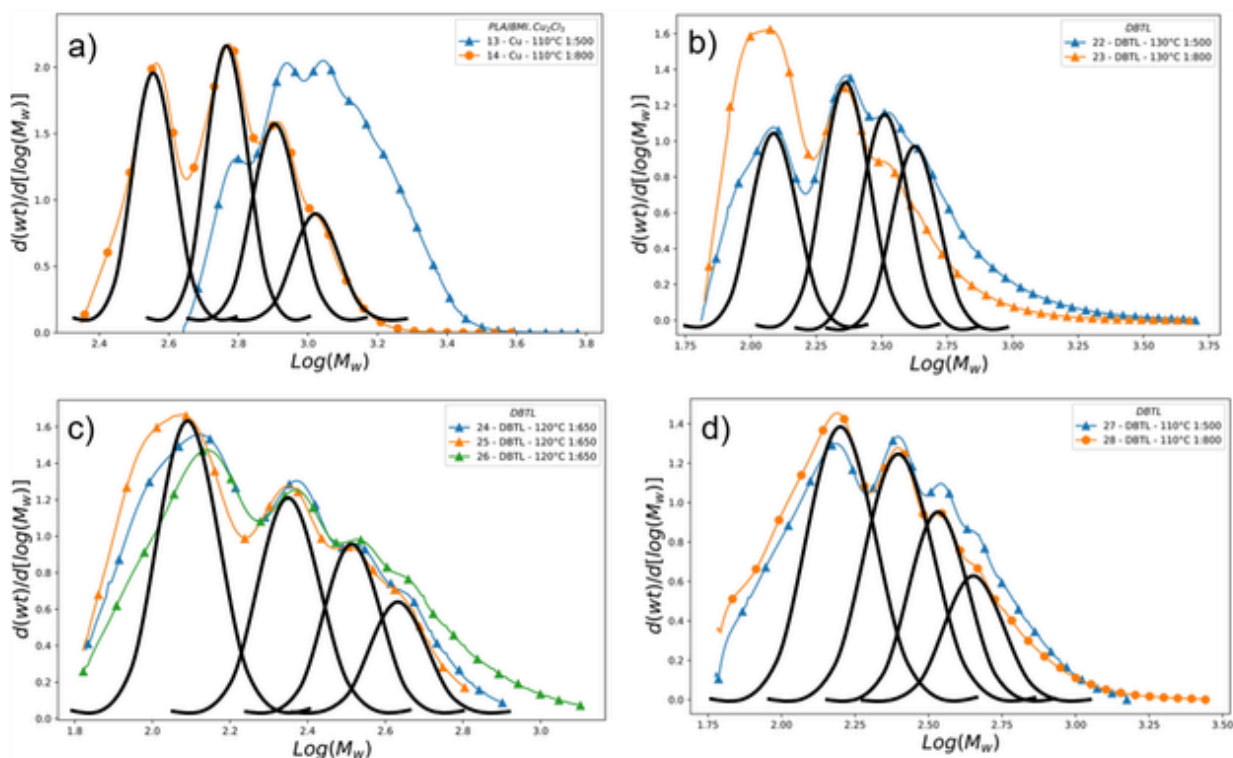


Figure X. Formation of micro molecules during ring-opening polymerization – (ROP) for producing poly (lactic acid) – (PLA).

The thermal behavior of polymers synthesized using ionic liquids different was carried out differential scanning calorimetry (DSC). The DSC curve (melting process) of 1 – In – 130°C 1:800, 16 – Sn – 130°C 1:800), 9 – Cu – 130°C 1:800, 23 – DBTL - 130°C 1:800 is presented in Figure 14. The thermal properties of these materials, such as melting temperature (T_m), melting enthalpy (H_m) and crystallinity (X_c) are reported in the Table 3. To calculate of crystallinity was used equation (1):

$$(Eq. 1)$$

where: ΔH_m is the melting enthalpy; ΔH_c° is the crystallization enthalpy; X_c is the crystallinity; and H_m° (J/g) is theoretical melting, corresponding the 93.7 J/g.

The transition temperature (T_g) was detected around 80-85°C, which is contrasted with some T_g value give in few papers, that it's reported around 65°C [20]. The melting temperature (T_m) was about 120 °C, which is reasonable in according to Vanaei et al. [21], even though the crystallinity to PLA synthesized using BMI.In2Cl7 and BMI.Sn2Cl5 was pretty low with values around 0.7% and 6.0%, respectively. The slow rate crystallization is co-related with the amorphous phase, due to the

low formation of lamellar fibrils during its crystal nucleation and, thus decreasing the mobility of polymer chains and preventing formation structures ordered [22, 23].

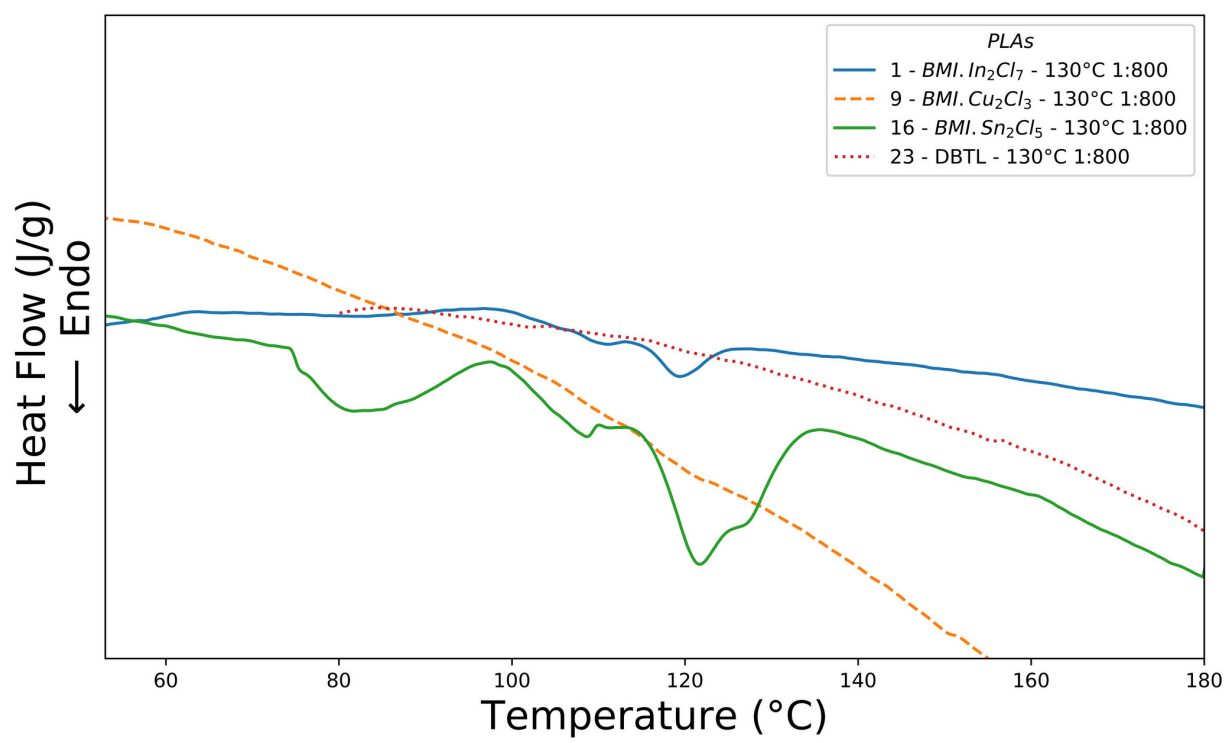


Figure 14. Time evolution of DSC of PLA at temperature 130°C - 1:800.

Table 3. melting temperature (T_m), melting temperature (H_m) and crystallinity (X_c) of the polymers synthesized at 130 °C and 1:800 using ILs and DBTL catalysts.

Sample	Catalyst	ΔH_m° (J/g)	ΔH_m (J/g)	T _g (°C)	T _m (°C)	X _c (%)
1	BMI.In ₂ Cl ₇	93.7	0.74	76	123	0.7
9	BMI.Cu ₂ Cl ₃	93.7	-	-	-	-
16	BMI.Sn ₂ Cl ₅	93.7	5.63	80	120	6.0
23	DBTL	93.7	-	-	-	-

Where, H°m theoretical melting enthalpy of PLA, 93.7 J/g [24].

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